

# Stat Chat

## Statistical Analysis Report

Generated: July 04, 2026 21:04  
Source: sample\_data.csv

### 1. Dataset Overview

|                             |     |
|-----------------------------|-----|
| Original rows               | 315 |
| Processed rows              | 280 |
| Rows removed                | 35  |
| Columns                     | 6   |
| Numeric columns             | 5   |
| Categorical cols            | 1   |
| Missing values (post-clean) | 0   |

Numeric columns: age, income, score, spend, label  
Categorical columns: region

### 2. Preprocessing Steps

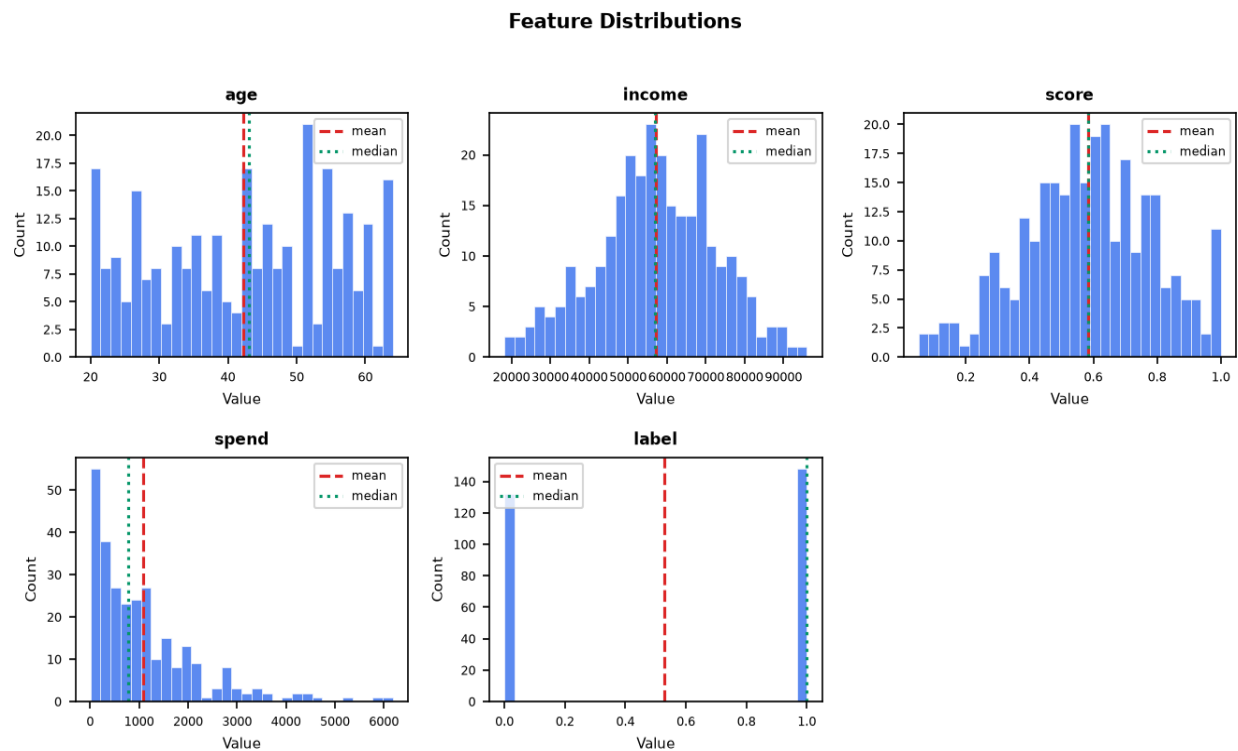
- Removed 15 duplicate row(s).
- Dropped 20 row(s) with null values.

### 3. Data Preview (first 10 rows — current version)

| age  | income     | score  | spend     | label | region |
|------|------------|--------|-----------|-------|--------|
| 58.0 | 61367.2937 | 0.6057 | 1075.2377 | 1     | West   |
| 48.0 | 69108.0338 | 0.7937 | 1178.5538 | 1     | North  |
| 34.0 | 41985.3837 | 0.4055 | 1891.772  | 1     | South  |
| 62.0 | 57187.9278 | 0.6159 | 287.2395  | 1     | West   |
| 27.0 | 34451.9341 | 0.9048 | 672.2177  | 1     | North  |
| 40.0 | 43423.2888 | 0.1087 | 250.0131  | 1     | North  |
| 58.0 | 68180.1032 | 0.4379 | 4395.5584 | 1     | East   |
| 38.0 | 51406.0824 | 0.4191 | 2883.4638 | 0     | East   |
| 42.0 | 73140.7296 | 0.5626 | 40.2251   | 0     | West   |

| age  | income     | score  | spend     | label | region |
|------|------------|--------|-----------|-------|--------|
| 30.0 | 63069.4001 | 0.5904 | 1222.7316 | 0     | South  |

## 4. Feature Distributions



## 5. Measures of Central Tendency

| Column | Mean       | Median     | Mode       | N   |
|--------|------------|------------|------------|-----|
| age    | 42.2821    | 43.0000    | 43.0000    | 280 |
| income | 57098.6971 | 56909.8437 | 18216.0310 | 280 |
| score  | 0.5828     | 0.5846     | 1.0000     | 280 |
| spend  | 1088.7865  | 779.3185   | 6.4063     | 280 |
| label  | 0.5286     | 1.0000     | 1.0000     | 280 |

## 6. Measures of Dispersion

| Column | Std Dev    | Variance       | Range      | IQR        | Min        | Max        |
|--------|------------|----------------|------------|------------|------------|------------|
| age    | 13.1466    | 172.8341       | 44.0000    | 23.0000    | 20.0000    | 64.0000    |
| income | 15149.9003 | 229519478.2554 | 77800.3014 | 20350.0015 | 18216.0310 | 96016.3324 |

|       |           |              |           |           |        |           |
|-------|-----------|--------------|-----------|-----------|--------|-----------|
| score | 0.2058    | 0.0424       | 0.9482    | 0.2762    | 0.0518 | 1.0000    |
| spend | 1087.4508 | 1182549.3113 | 6175.7881 | 1241.3997 | 6.4063 | 6182.1944 |
| label | 0.5001    | 0.2501       | 1.0000    | 1.0000    | 0.0000 | 1.0000    |

## 7. Distribution Shape (Skewness & Kurtosis)

| Column | Skewness | Kurtosis | Skew Interpretation |
|--------|----------|----------|---------------------|
| age    | -0.0990  | -1.1914  | Symmetric (left)    |
| income | -0.1300  | -0.2649  | Symmetric (left)    |
| score  | -0.1199  | -0.2922  | Symmetric (left)    |
| spend  | 1.7627   | 3.8101   | High skew (right)   |
| label  | -0.1151  | -2.0011  | Symmetric (left)    |

## 8. Percentile Statistics

| Column | P5         | P10        | P25        | P50        | P75        | P90        | P95        |
|--------|------------|------------|------------|------------|------------|------------|------------|
| age    | 21.0000    | 23.0000    | 31.0000    | 43.0000    | 54.0000    | 60.0000    | 63.0000    |
| income | 30777.2410 | 35778.6699 | 47805.3327 | 56909.8437 | 68155.3342 | 76677.3991 | 81590.7150 |
| score  | 0.2474     | 0.2991     | 0.4427     | 0.5846     | 0.7189     | 0.8499     | 0.9242     |
| spend  | 37.5511    | 84.2513    | 271.1371   | 779.3185   | 1512.5369  | 2596.9907  | 3304.1943  |
| label  | 0.0000     | 0.0000     | 0.0000     | 1.0000     | 1.0000     | 1.0000     | 1.0000     |

## 9. Normality Tests (Shapiro-Wilk)

| Column | Statistic | p-value | Normal (p>0.05)? |
|--------|-----------|---------|------------------|
| age    | 0.9503    | 0.0000  | No               |
| income | 0.9948    | 0.4600  | Yes              |
| score  | 0.9916    | 0.1135  | Yes              |
| spend  | 0.8301    | 0.0000  | No               |
| label  | 0.6353    | 0.0000  | No               |